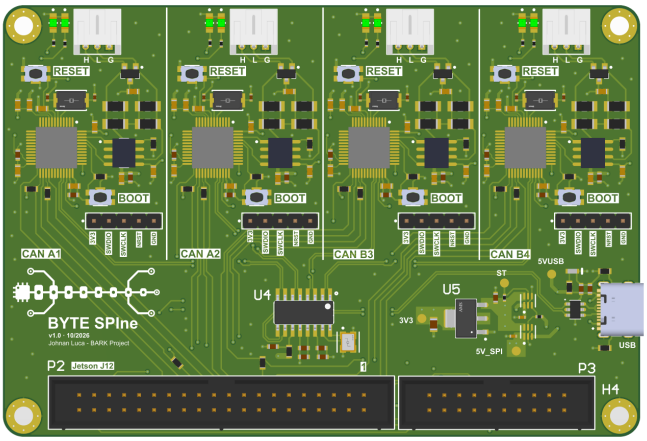


Impedance and Signal Integrity Simulation of the "BYTE SPIne" SPI-to-CAN Bridge

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BYTE SPIne version 1.0

Johnan Luca
BARK Project - AI Team-Robotics

Companion document to *Architecture Design and Sizing of the BYTE SPIne* (`BYTE_SPIne_Documentation.pdf`). It records every setting, step and problem of the simulations, so that the results can be reproduced.

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1 Scope and Summary

This document describes two simulations run in Altium Designer 26 on the routed PCB of the BYTE SPIne:

1. **Differential impedance** of the USB pairs, with the field solver of the Layer Stack Manager (Section 2);
2. **Signal integrity** of the SPI1 bus (Jetson → MCU B3/B4), with the built-in *Signal Integrity* analyser [6] and IBIS models of the real drivers and receivers (Section 3).

Both tools are included in the standard Altium Designer licence. The DC voltage-drop analysis (*Power Analyzer by Keysight*) is a paid add-on and was not used: the hand calculation of the main documentation [1] (Section 9.6) already shows a drop of only 5 mV to 11 mV on the 5 V tree.

For each simulation the document gives the theory needed to read the result, the exact settings, the step-by-step procedure, the problems found on the way with their solution, and the results. Settings and results are given as tables and as plots generated from data exported from Altium, not as screenshots. Section 4 lists every change made to the project for the simulations, and Section 5 the work still open.

Test	Result	Verdict
USB impedance	90.18 Ω differential on L1 and L4 (target 90 Ω ±10 %); hand formula 92 Ω	pass
SPI1 MISO	clean driver thanks to the 33 Ω series resistor R4; margin 0.43 V at the Jetson	pass
SPI1 MOSI	ringing re-enters the threshold band after the rising edge (down to 1.60 V), leaves it 20 ns after the edge	acceptable (data line)
SPI1 SCK	first ring down to 2.09 V, 0.22 V below the guaranteed V_{IH}	marginal
SPI1 SCK, sweep 10–56 Ω	first ring at 2.54 V to 3.12 V	pass
SPI1 SCK, 33 Ω on the layout	first ring at 2.79 V (B4) and 2.96 V (B3): margin ≥0.48 V	pass

Table 1: Summary. Design decision, implemented on the layout: one 33 Ω series resistor on each SCK line, close to P2 (Section 3.8).

2 Test 1: Differential Impedance of the USB Pairs

2.1 Theory

The characteristic impedance of a track depends only on its cross-section: track width w, copper thickness t, height h and relative permittivity ε_r of the dielectric between the track and its reference plane, and, for a differential pair, the gap s between the two lines. For an edge-coupled microstrip (outer layer over a plane) IPC-2141 gives the closed-form approximation

$$Z_0 \approx \frac{87}{\sqrt{\varepsilon_r + 1.41}} \ln\!\left(\frac{5.98\,h}{0.8\,w + t}\right), \qquad Z_{\text{diff}} \approx 2Z_0\left(1 - 0.48\,e^{-0.96\,s/h}\right), \tag{1}$$

valid for $0.1 < w/h < 2$ [5]. With $w = 0.25$ mm, $s = 0.15$ mm, $t = 35$ µm, $h = 0.2104$ mm and $\varepsilon_r = 4.4$ ($w/h = 1.19$) it yields $Z_0 \approx 60.6\,\Omega$ and $Z_{\text{diff}} \approx 91.8\,\Omega$, without solder mask. A 2D field solver computes the electromagnetic field of the real cross-section, including the solder

mask that covers the tracks, and therefore gives the exact value. If the two methods agree, both the geometry and the stack-up entered in the tool are correct.

2.2 Stack-up Check

Design » Layer Stack Manager, tab Stackup [7]. The layers must match the JLCPCB stack-up JLC04161H-7628, including the two solder-mask layers (15.24 µm, $\varepsilon_r = 3.8$), which lower the impedance by a few ohms. The stack-up entered in the design is shown in the Draftsman export of Figure 1: prepreg 0.2104 mm ($\varepsilon_r = 4.4$), core 1.065 mm ($\varepsilon_r = 4.43$, value of the NP-155F library material), outer copper 1 oz, inner copper 0.5 oz.

2.3 Procedure

1. Tab Impedance (bottom of the Layer Stack Manager), + Add. The first profile created is single-ended 50 Ω by default (kept as *S50* for reference: 0.358 mm on the outer layers).
2. + Add again and open the Properties panel (Panels » Properties; it is closed by default): Type **Differential**, Target Impedance **90**, Target Tolerance **10 %**, Description *USB 90R*.
3. The profile table shows one row per signal layer with its reference planes. Altium keeps the gap at the default 0.127 mm and computes the width (0.224 mm for 90 Ω).
4. In the rows **Top Layer** (Bottom Ref = L2 GND) and **Bottom Layer** (Top Ref = L3 PWR) type the routed geometry: Trace Gap **0.15**, then Width **0.25**. The column *Impedance* now shows the impedance of the routed pairs and *Deviation* the error with respect to the target.

Pitfall: the two middle rows (L2 and L3) are inner layers: a track there is a stripline between two planes. Entering 0.25/0.15 on these rows by mistake gives 79.4 Ω (Table 2), which is irrelevant because no USB track is routed on the inner layers. Those rows were reset to the gap of 0.127 mm so that the profile stays clean.

Step	Rows	Width	Gap	Impedance
Profile as created	all rows (width solved by Altium)	0.224 mm	0.127 mm	90 Ω
Wrong rows edited	L2, L3 (inner layers, stripline)	0.25 mm	0.15 mm	79.4 Ω (−11.7 %)
Final	**L1 and L4** (routed geometry)	0.25 mm	0.15 mm	90.18 Ω (+0.2 %)

Table 2: States of the differential profile *USB 90R* (Type Differential, target 90 Ω, tolerance 10 %). In the final profile the inner rows are back at the default gap; the complete table is the Draftsman export of Figure 1.

2.4 Result

The field solver gives 90.18 Ω for the routed pairs on both outer layers. The hand formula gives 91.8 Ω without solder mask; the difference of 1.8 % is the effect of the solder mask. The profile was not linked to the differential-pair routing rule (Use Impedance Profile), because the rule already uses 0.25/0.15 and the DRC is clean. The JLCPCB impedance calculator [9] (JLC04161H-7628, differential microstrip, L1 referenced to L2) can be used as a third, independent check.

2.5 Export of the Impedance Table (Draftsman)

The table is documented with a Draftsman sheet generated from the PCB, which is the output of the tool itself and not a table typed by hand (Figure 1):

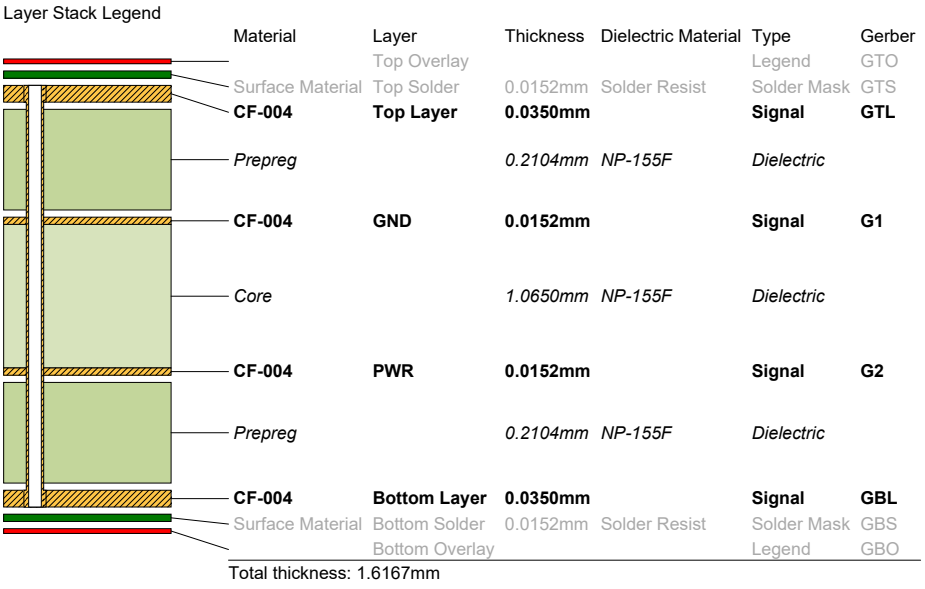
1. In the Projects panel: right-click on the project, Add New to Project » Draftsman Document. No A4 template is installed; *ISO A3 Landscape* or the default sheet is sufficient, since only the tables are used.

2. Place » Layer Stack Legend (in AD26 the layer stack table is part of the legend) and Place » Transmission Line Table [8].

3. Transmission Line Table: in its Properties, hide the columns *Impedance Id*, *Narrow Trace Width* (equal to the width, no etch factor), *Clearance* (no coplanar ground) and *Substrate*.

4. Layer Stack Legend: the thicknesses are rounded to two decimals (0.035 mm shown as 0.04) and the units of the legend itself cannot be edited. The precision is set at document level: click on an empty area, Properties » Units, *Value Precision* = **0.0001**; then delete and place the legend again.

5. File » Export to PDF, saved as `Documentation\BYTE_SPIne_Stackup.pdf`.



Transmission Line Structure Table

Transmission Line	Target Impedance	Calculated Impedance	Trace layer	Width	Gap	Reference layers	Target Tolerance
Edge-Coupled Coated Microstrip	90	90.18	Top Layer	0.2500mm	0.1500mm	GND	10%
Edge-Coupled Offset Stripline	90	90.03	GND	0.1578mm	0.1270mm	Top Layer,PWR	10%
Edge-Coupled Offset Stripline	90	90.03	PWR	0.1578mm	0.1270mm	GND,Bottom Layer	10%
Edge-Coupled Coated Microstrip	90	90.18	Bottom Layer	0.2500mm	0.1500mm	PWR	10%

Figure 1: Draftsman export: layer stack and differential impedance profile.

3 Test 2: Signal Integrity of the SPI1 Bus

3.1 Motivation and Topology

The first version of the main documentation [1] assumed SPI edges of 5 ns to 10 ns for all lines, hence a critical length above 130 mm and no need for any termination. SPI1 MOSI is the longest

line of the board (112 mm), close to that limit, so the assumption was checked with a simulation. The simulation showed that it does not hold for the lines driven by the Jetson (Section 3.7); the critical lengths of the main documentation have since been corrected (Table 11 there). The topology of each SPI1 line is a **star at the pads of P2**: one branch on the top layer to MCU B3, one branch (via to the bottom layer) to MCU B4.

- SCK, MOSI, NSS3, NSS4 are driven by the Jetson and received by the two MCUs;
- MISO is driven by the selected MCU through its 33 Ω series resistor R4 and received by the Jetson.

On the Jetson Orin Nano developer kit the 40-pin header is driven through **TXB0108** level translators (1.8 V SoC side, port A; 3.3 V header side, port B). The real driver of SCK/MOSI/NSS is therefore the port B of a TXB0108, not the SoC.

3.2 Acceptance Criteria

A line passes when, at every receiver, the edge crosses the input threshold band **once** (monotonic crossing) and the ringing after the edge does not re-enter the band:

Receiver	V_{IL}	V_{IH}	Source
STM32F072, FT pin, $V_{DD} = 3.3$ V	0.99 V	2.31 V	IBIS `io8p.._sudq_ft`
TXB0108 port B, $V_{CCB} = 3.3$ V	1.05 V	2.34 V	IBIS `TXB0108B_IOB_33`

Table 3: Input thresholds used as criteria (guaranteed values, 0.3/0.7 V_{DD}).

On a **clock** line a re-entry into the band can be seen as an extra edge (one bit too many). On a **data** line it is harmless if the ringing has died out before the sampling edge of the clock (125 ns to 500 ns after the data edge at 1 MHz to 4 MHz). The STM32 inputs have a Schmitt trigger; the typical switching points (about 1.3 V to 1.6 V) are inside the band, which gives practical but not guaranteed margin.

3.3 IBIS Models

STM32F072C8T6. Downloaded from st.com [2] (product page, *CAD Resources*, *IBIS models*) into `SI_Models\stm32f0_ibis`. The archive contains one file per silicon family and package; the number in the file name is the **DEV_ID** of the family:

DEV_ID	Family
0x440	STM32F030x8, STM32F05x
0x442	STM32F09x, STM32F030xC
0x444	STM32F03x (F030x4/x6, F031)
0x445	STM32F04x, STM32F070x6
0x448	**STM32F070xB, STM32F071, STM32F072**

Table 4: The file to use is `stm32f0_448_lqfp48.ibs`.

Required: import `stm32f0_448_lqfp48.ibs`: **DEV_ID 0x448** is the silicon of the STM32F072. The other files of the archive belong to other STM32F0 families and give wrong buffers.

The file defines, for each pin type, a *model selector* with several variants. For the FT pins (PB12–PB15):

Name part	Meaning (read from the file)
`io8p01`, `io8p10`, `io8p11`	GPIO speed. Rise ramps ≈12.5 ns, 6.4 ns, 1.5 ns: `p01` is the slowest, i.e. **Low speed** as configured in CubeMX
`_lv`	supply 2.0 V (thresholds 0.72/1.68 V)
no `_lv`	supply 3.3 V (thresholds 0.99/2.31 V)
`_pd`, `_pu`	internal pull-down / pull-up (40 kΩ) enabled

Table 5: Model variants of the STM32 IBIS file.

When importing, Altium converts each model into a `.mac` file and appends `_tp` (typical corner) and `_in`/`_out` (input/output part of the buffer): `io8p01_sudq_ft` becomes `io8p01_sudq_ft_tp_in.mac` and `io8p01_sudq_ft_tp_out.mac`.

TXB0108. Downloaded from ti.com [3] into `SI_Models\TXB0108_IBIS\txb0108.ibs` (component `TXB0108YZP_PW`). Port B is available at 1.8, 2.5, 3.3 and 5 V, with and without the *one-shot* edge accelerator:

Model	Behaviour	Edge (≈2 V)	C_{comp}
`TXB0108B_IOB_33`	buffer without accelerator	0.6 ns	9.4 pF
`TXB0108_IOB_OS_33`	one-shot active	1.4 ns to 2.5 ns	9.2 pF

Table 6: TXB0108 port B models at 3.3 V.

The TXB0108 [4] has a weak output (about 4 kΩ) and a one-shot that drives the line hard for a few nanoseconds on every edge. TI states in the IBIS file that this change of impedance during the transition cannot be described by IBIS and that the results may not be accurate. The simulations of this document use `TXB0108B_IOB_33` (fastest edge), i.e. the **worst case**: a line that passes with this model also passes with the real device.

3.4 Procedure

3.4.1 Start and Model Assignments

1. Project » Validate PCB Project. From the PCB: Tools » Signal Integrity. The warning *Not all components have Signal Integrity models set up* appears: Model Assignments....

2. The table lists one entry per logical component (multi-channel: the model of U1 applies to the four MCUs). Passives (*Low Confidence*) are modelled from their value and are correct. Components with *No Match* that are not on the analysed nets (H1–H4, TP1–TP4, Y1, Y2, P3, D1, S1, S2) are ignored by the analyser and left as they are.

   Errors or warnings found — Not all components have Signal Integrity models set up. [Model Assignments...] [Continue] [Cancel]

This warning appears at every start, because some components on other nets have no model; Continue is correct once U1 and P2 are set.

3.4.2 U1 (STM32)

1. Double-click on U1, Model Name `STM32F072C8_LQFP48`, **Technology LVC** (see Section 3.5), Import IBIS, file `stm32f0_448_lqfp48.ibs`. The models are mapped to the pins by number (pin 44 BOOT0 receives `vpp_test`, the I^2C pins the `i2c` models: the mapping is correct).

2. The automatic choice is `io8p01_sudq_ft_pd_lv`: right speed, but 2.0 V supply and internal pull-down, both wrong for this design. On the SPI2 pins set:

Pin	Signal	Input model	Output model
25	PB12, NSS	`io8p11_sudq_ft_tp_in`	`io8p11_sudq_ft_tp_out`
26	PB13, SCK	`io8p11_sudq_ft_tp_in`	`io8p11_sudq_ft_tp_out`
27	PB14, MISO	`io8p01_sudq_ft_tp_in`	`io8p01_sudq_ft_tp_out` (Low speed)
28	PB15, MOSI	`io8p11_sudq_ft_tp_in`	`io8p11_sudq_ft_tp_out`

Table 7: U1 pin models. On the inputs only the input part is used, which is identical for all speeds (same thresholds, $C_{\text{comp}} = 1.97\,\text{pF}$); the speed matters only on MISO, the only pin driven by the MCU.

Required: set **Technology = LVC** in the component field of **U1 and P2** (not in the per-pin column, which stays on *Default*). With the default *HC* every buffer is supplied at 5 V and all waveforms are wrong (Section 3.5).

3.4.3 P2 (Jetson Header)

1. Double-click on P2, Type **IC**, Model Name `P2_Jetson_TXB0108`, **Technology LVC**, Import IBIS, `txb0108.ibs`, component `TXB0108YZP_PW`. The 20 pins of the TXB0108 are mapped by number onto pins 1–20 of P2, i.e. at random with respect to the board; only the analysed pins are corrected.

2. **Problem:** the models cannot be edited, because all pins of P2 are *Passive* (electrical type taken from the schematic). **Solution:** in the schematic select P2, Properties » Pins, pencil button (*Component Pin Editor*), set *Type* = **I/O** on pins 13, 16, 18, 22 and 37. P2 comes from Manufacturer Part Search, so this is a local change of the placed component; it does not affect the PCB nor the ERC.

3. Validate the project, reopen Model Assignments and set on P2:

Pin	Signal	Input model	Output model
13	SPI1_SCK	`TXB0108B_IOB_33_tp_in`	`TXB0108B_IOB_33_tp_out`
37	SPI1_MOSI	`TXB0108B_IOB_33_tp_in`	`TXB0108B_IOB_33_tp_out`
18	SPI_NSS3	`TXB0108B_IOB_33_tp_in`	`TXB0108B_IOB_33_tp_out`
16	SPI_NSS4	`TXB0108B_IOB_33_tp_in`	`TXB0108B_IOB_33_tp_out`
22	SPI1_MISO	`TXB0108B_IOB_33_tp_in`	`TXB0108B_IOB_33_tp_out`

Table 8: P2 pin models (3.3 V port B). The automatic import had assigned the 1.2 V/1.8 V variants (first entries of the selectors); pins 13 and 16 had received the GND and OE models of the TXB0108.

Required: the analysed pins of P2 (13, 16, 18, 22, 37) must have the electrical type **I/O** in the schematic; with *Passive* pins Altium does not let the models be edited.

3.4.4 Writing the Models to the Schematic

At Analyze Design Altium warns that *some unsaved components have pin parameters* which are not used unless written to the schematic. The per-pin models are such parameters: answer No,

tick *Update Schematic* on the rows **U1** and **P2**, click Update Models in Schematic, save, and analyse again. The models become SI parameters of the two components; they have no effect on PCB, DRC or BOM.

Required: after every change of the models, click **Update Models in Schematic** for U1 and P2 and save; otherwise the per-pin models are ignored by the analysis.

> *Some unsaved components have pin parameters. If these components have not been updated on the schematic, any associated pin parameters will not be used during the analysis. Continue Anyway?* [Yes] [No]

3.4.5 Analysis Setup and Rules

The dialog *SI Setup Options* applies only to nets that are not routed; all nets are routed, so the real tracks and the stack-up of Section 2 are used and the defaults can be kept:

Setting	Value	Used for
Track Impedance	100 Ω	nets not yet transferred to the PCB
Average Track Length	2000 mil	nets placed but not routed
Use Manhattan length	enabled	nets placed but not routed

Table 9: SI Setup Options (defaults, not used for the routed SPI nets).

The other settings are PCB rules (Design » Rules » Signal Integrity). The table below is generated from the rule export of the PCB (Design » Rules, right-click on *Signal Integrity*, Export Rules, file `SI_Data\BYTE_SPIne.RUL`):

Rule (name in the export)	Value	Reason

`SupplyNets_5VUSB`
`SupplyNets_5V_SPI`
`SupplyNets_5V`
`SupplyNets_3V3_Jetson`
`SupplyNets_3V3`
`SupplyNets_GND`
	5.000 V
	5.000 V
	5.000 V
	3.300 V
	3.300 V
	0.000 V
		one rule per power net. Without them R3 (NSS pull-up) and the clamps of the buffers have no reference: 3V3 and 5V were flagged red and the NSS nets were *Not analyzed*

`SignalStimulus`	single pulse, low start, 10 ns to 60 ns, period 100 ns	default; only the edges matter for reflections
`SignalTopValue`	min. 3.300 V	pass criterion for the high level (default 5 V)
Signal Base Value	**deleted**	with a maximum of 0 V every net fails on a few microvolts; without the rule the base level is not checked, which is acceptable because the waveforms are judged directly

Table 10: Signal integrity rules (from `BYTE_SPIne.RUL`). The export also contains the rule *Schematic Supply Nets* (GND, 0 V), created automatically by Altium from the power ports of the schematic. The scope of each Supply Nets rule is `InNet('<net>')`, the scope of the other rules is *All*.

Required: define **one Supply Nets rule per power net** and **delete the Signal Base Value rule**; without the first the pull-ups and clamps have no reference, with the second almost every net fails on a meaningless criterion.

3.4.6 Running a Reflection Analysis

1. In the Signal Integrity panel, double-click on the net (or select it and click >): it moves to the list on the right and its pins are listed below.
2. Right-click on each pin and set the direction: the driver **Out**, the receivers **In** (Table 11).
3. Reflection Waveforms.... The result opens as a separate document `BYTE_SPIne.sdf`, one tab per run.
4. For a termination sweep tick **Serial Res** in the Termination list (*No Termination* cannot be unticked: it is always simulated as reference), keep *Perform Sweep* with 10 steps; the series resistor is placed at the driver.

Net	Out	In
SPI1_MOSI	P2-37	U1_CAN_B3-28, U1_CAN_B4-28
SPI1_SCK	P2-13	U1_CAN_B3-26, U1_CAN_B4-26
SPI1_MISO	U1_CAN_B4-27 (farthest MCU)	P2-22, U1_CAN_B3-27 (high impedance)

Table 11: Pin directions.

Required: set the directions of every analysed net before each run: the driver **Out**, the receivers **In**. With inverted directions the waveforms describe a different circuit (Section 3.5).

3.5 Problems Found and Solutions

Symptom	Cause	Solution
P2 models cannot be edited	P2 pins are Passive in the schematic	electrical type I/O on the SPI1 pins (Component Pin Editor)
Warning about pin parameters	per-pin models are used only if written to the schematic	Update Models in Schematic for U1 and P2
3V3/5V red, NSS nets *Not analyzed*	no Supply Nets rule	one Supply Nets rule per power net
Reflection Waveforms does nothing	no net in the list on the right	move the net with > or double-click
All waveforms swing 0 V to 5 V, Top Value 5.000 V on every net	component Technology *HC* (74HC family, 5 V)	Technology **LVC** on U1 and P2 (Section 3.5)
Almost every net fails after changing the rules	Base Value maximum 0 V; overshoot measured against the Top Value (3.3 V), so nets still at 5 V show ≈1.7 V of "overshoot"	Signal Base Value rule deleted; judge the result on the waveforms, not on the screening table
MISO waveform unrealistic	directions inverted (P2 set as driver)	U1_CAN_B4-27 Out, P2-22 In
Screening and waveforms disagree (e.g. MOSI overshoot 2.53 V in the table, ≈0.7 V on the waveform)	the screening is a fast approximate calculation	use the screening only to choose the nets to simulate

Table 12: Problems met during the setup.

Screening of the final layout. The screening table of the Signal Integrity panel is exported with a right-click on the net list, Display Report; Altium writes `Signal Integrity Tests Report.txt` in the project folder (copy in `SI_Data\si_screening_report.txt`). The report

lists, for each criterion, only the nets outside the tolerance (0.66 V for the SPI1 nets). The SPI1 rows:

Net	Falling edge overshoot	Falling edge undershoot	Rising edge overshoot	Rising edge undershoot	Top Value
SPI1_MOSI	2.513 V	1.637 V	2.361 V	1.484 V	pass
SPI1_SCK (MCU side)	1.407 V	pass	1.240 V	pass	pass
NetP2_13 (P2 side)	1.407 V	pass	1.240 V	pass	pass
SPI1_MISO, NetR4_CAN_B3/B4_1	pass	pass	pass	pass	3.292 V

Table 13: Screening of the SPI1 nets on the final layout (`si_screening_report.txt`). "pass" means that the net is not listed for that criterion. SPI1_SCK and NetP2_13 are the two sides of R17.

The Top Value of the SPI1 nets is 3.3 V (MISO 3.292 V, below the minimum of 3.3 V only because of the drop on R4), which confirms that the 3.3 V configuration of Section 3.5 is active. The overshoot values, instead, do not match the waveforms (for example 2.5 V for MOSI) and do not change with R17 on SCK: the screening is a fast approximate calculation, used here only to choose the nets to simulate.

The 5 V swing. The first waveforms swung from 0 to 5 V on driver and receivers, although all models are 3.3 V models: on SPI1_MOSI the driver P2-37 settled at 5.0 V and the receivers peaked at about 5.5 V (B3) and 5.7 V (B4). The shape of the ringing was correct, only the amplitude was wrong. Two attempts had no effect: setting the Supply Nets rule of 5V_SPI (the 5 V pins of P2) to 3.3 V, and setting the Signal Top Value rule to 3.3 V (it is only a pass criterion). The cause is the field **Technology** of the component in the model dialog, which defaults to *HC* (74HC logic, 5 V) and sets the supply of the buffers. Setting it to **LVC** (3.3 V CMOS) on U1 and P2, writing the models to the schematic and re-analysing gives the correct 3.3 V swing. Only the component field must be changed: the per-pin Technology column stays on *Default*, otherwise the generic LVC models could replace the IBIS models.

Required: when reproducing the analysis, check the column **Top Value** of the SPI1 nets (≈3.3 V) **before reading any waveform**.

The 5V_SPI rule was set back to 5 V. The Technology list of the model dialog offers the logic families of the generic models (GTL_LVT, HC, HCT, HLL, LS, LV, LVC, ...); LVC is the 3.3 V CMOS family.

Inverted MISO directions. In a first MISO run P2 was set as driver by mistake. With the fast TXB0108 edge driving the branched line, the MCU pins showed a ringing up to about 4.3 V (B3) and 4.7 V (B4) and an undershoot of about −0.8 V. This is not the real case (MISO is driven by the MCU through R4), but it shows the same effect that is found later on SCK and MOSI.

3.6 Results

The waveforms were exported from the SimData document (click in a plot, File » Export » Plot, CSV) and plotted with `plot_si.py` (Section 4); all values below are computed from the exported data. The threshold band of each receiver is shaded. Definitions:

- *minimum after rise*: lowest point of the ringing after the first overshoot peak (a shoulder on the edge itself, above V_{IH}, is not a re-entry);
- *maximum after fall*: highest point of the ringing after the first undershoot;

- *margin*: minimum after rise $-V_{IH}$, and $V_{IL}-$ maximum after fall.

Net	Receiver	Peak	Min. after rise	Max. after fall	Under-shoot	Margin high / low	Verdict
MISO	P2-22 (Jetson)	4.51	2.77	0.28	−0.75	+0.43 / +0.77	pass
MISO	B3-27	4.60	2.72	0.32	−0.78	+0.41 / +0.67	pass
MOSI	B3-28	5.04	1.85	0.62	−0.80	−0.46 / +0.37	acceptable (data)
MOSI	B4-28	5.31	1.60	0.63	−0.80	−0.71 / +0.36	acceptable (data)
SCK, no R	B3-26	4.79	2.18	0.55	−0.79	−0.13 / +0.44	marginal
SCK, no R	B4-26	4.91	2.09	0.53	−0.78	−0.22 / +0.46	marginal
SCK, 33 Ω	B3-26	3.71	2.96	0.28	−0.29	+0.65 / +0.71	**pass**
SCK, 33 Ω	B4-26	4.10	2.79	0.42	−0.76	+0.48 / +0.57	**pass**

Table 14: Results in volts (worst-case driver model). Thresholds: STM32 0.99/2.31 V, TXB0108 1.05/2.34 V. "SCK, 33 Ω" is the analysis of the final layout, with the resistor placed and routed.

MISO. The driver waveform at B4 is clean and shows the typical step of a source-terminated line: R4 (33 Ω) works as intended. The remaining ringing at the Jetson comes from the star topology, where the branch to B3 acts as an unterminated stub; it stays out of the threshold band with 0.43 V of margin. The overshoot (4.5 V) is within the absolute maximum of the TXB0108 inputs (6.5 V); the undershoot of −0.75 V for about 2 ns slightly exceeds the −0.5 V rating, with the current limited by R4, and will be reduced on the real board by the capacitance of the Jetson carrier, which is not modelled.

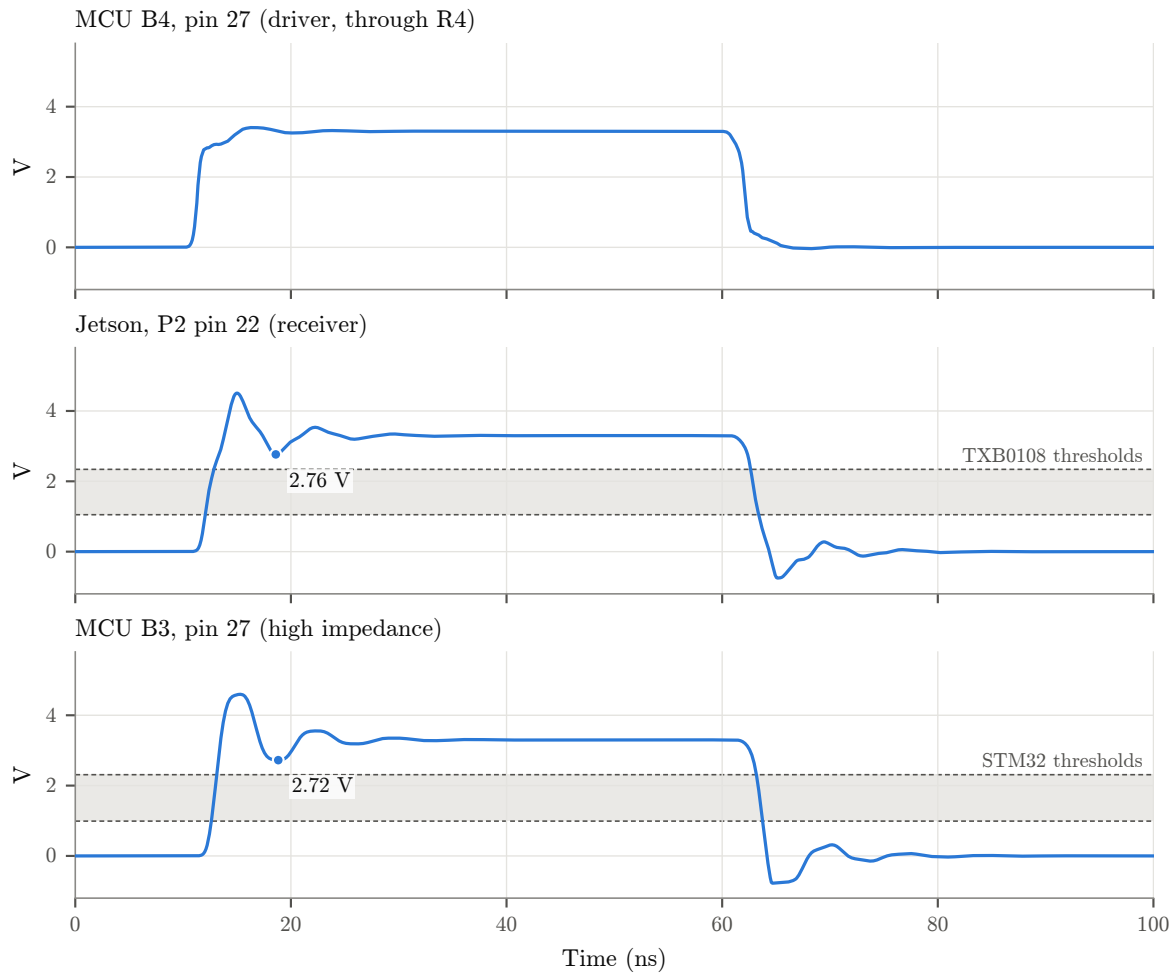


Figure 2: SPI1_MISO, MCU B4 driving through R4.

MOSI. The ringing after the rising edge re-enters the threshold band at both MCUs, by up to 0.71 V at B4. MOSI is a data line sampled on the clock edge, 125 ns to 500 ns after the data edge at 1 MHz to 4 MHz, while the ringing has left the band about 20 ns after the edge: the value is stable when it is sampled. Acceptable without termination.

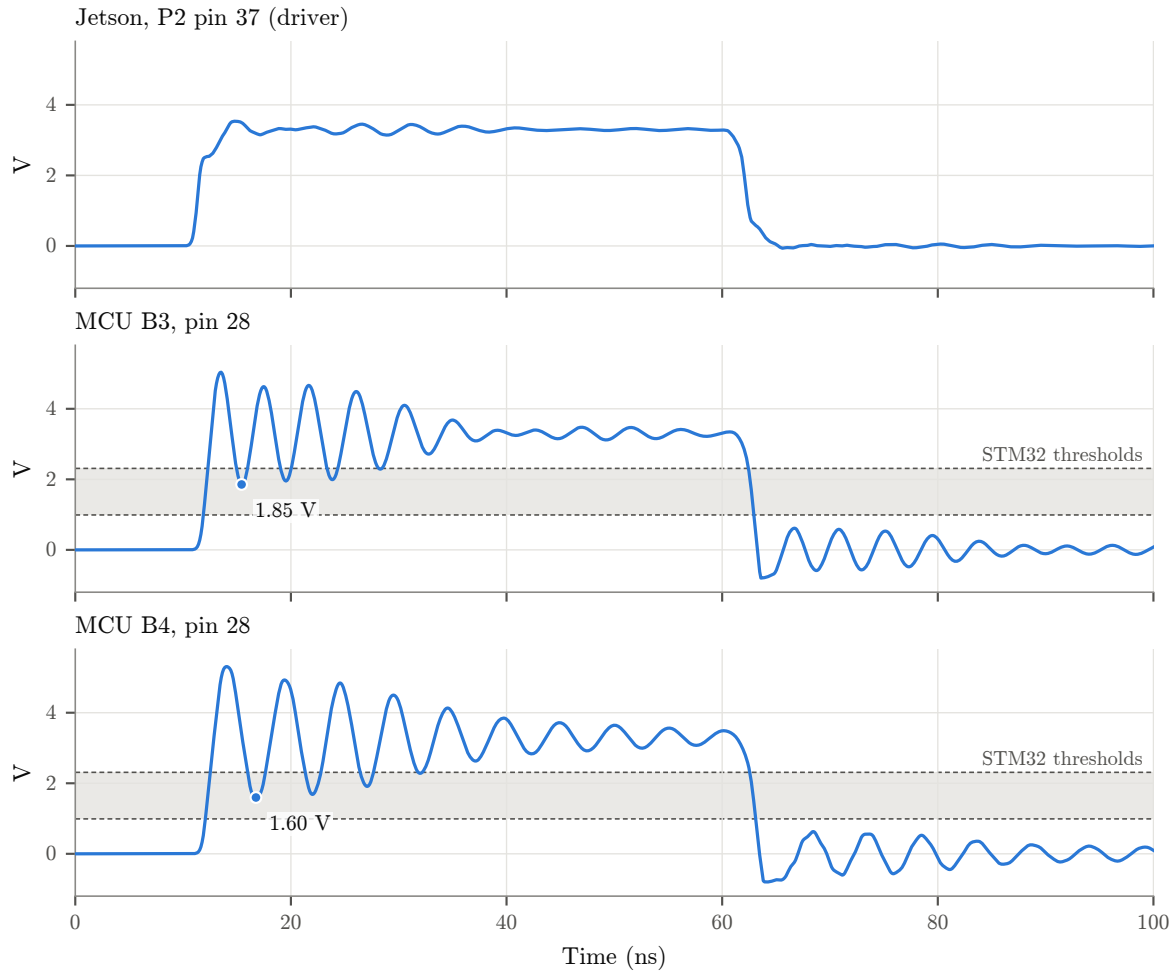


Figure 3: SPI1_MOSI, Jetson driving.

SCK. Without series resistor the first ring after the rising edge falls to 2.09 V at B4 and 2.18 V at B3, i.e. 0.13 V to 0.22 V below the guaranteed V_{IH}, and the line keeps ringing during the whole pulse (resonance of the star). It stays far above V_{IL} and above the typical switching point of the Schmitt trigger, so a double clock is unlikely, but the margin with respect to the datasheet is negative on the line where an extra edge costs a bit. With the 33 Ω resistor placed and routed on the layout (Figure 4, blue) the first ring stays at 2.79 V (B4) and 2.96 V (B3): margin 0.48 V or more, peak reduced from 4.9 to 4.1 V. The driver waveform at P2 is practically unchanged.

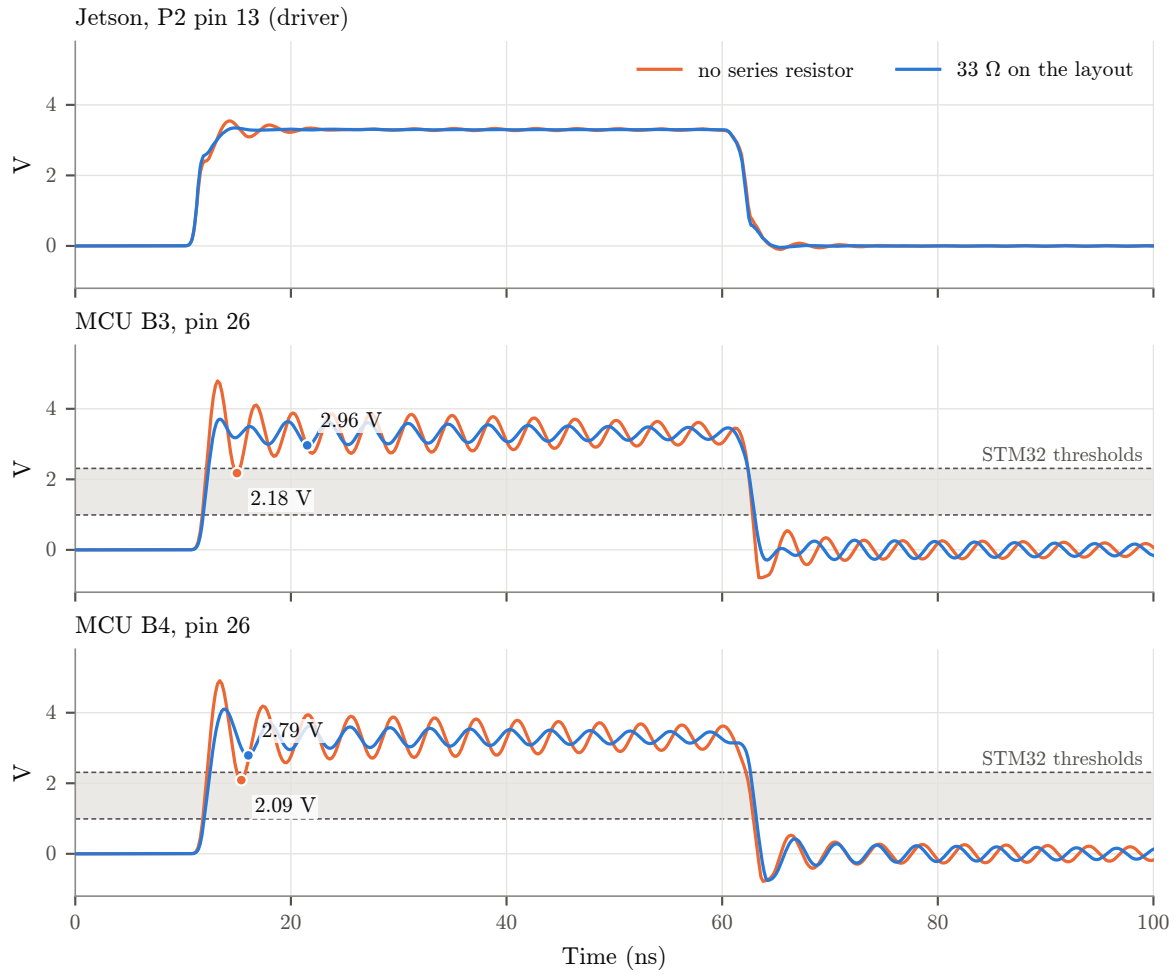


Figure 4: SPI1_SCK before (orange) and after (blue) the 33 Ω series resistor, analysis of the final layout. The marked points are the minima after the rising edge.

Series-resistor sweep. Before the layout change, the sweep of the Signal Integrity panel (*Serial Res*, 10 to 56 Ω in 10 linear steps of 5.1 Ω, resistor ideally placed at the driver) was used to choose the value. The minimum after rise at B4 grows monotonically from 2.54 V (10 Ω) to 3.12 V (56 Ω), and the peak falls from 4.52 to 3.53 V: every value of the range removes the re-entry. Larger values also slow down the edge, so a value in the middle of the range is the right compromise. The layout result with 33 Ω (2.79 V) is about 0.15 V lower than the sweep at the same value (2.95 V to 3.00 V), because on the board the resistor sits a few millimetres from the pad and the bottom branch starts after a via; the margin is still ample.

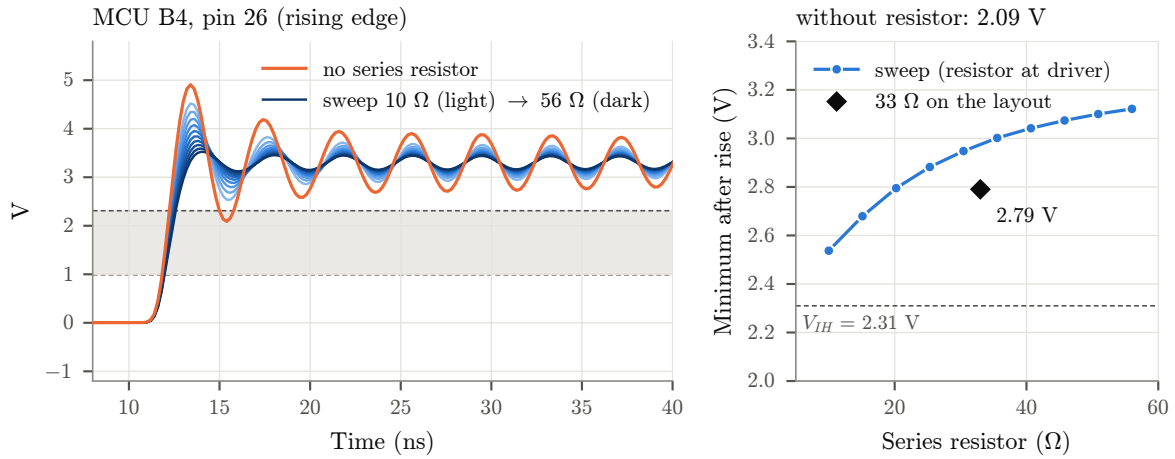


Figure 5: SPI1_SCK at MCU B4: rising edge for the ten values of the sweep (left) and minimum after rise as a function of the series resistor (right), with the result of the final layout.

3.7 Interpretation and Limits

- **The assumption of the first version of the main documentation is not valid for the Jetson-driven lines.** The STM32 at low speed has edges of ≈12 ns (MISO is clean), but the TXB0108 drives edges of 0.6 ns to 2.5 ns. With 6.5 ps/mm and the one-sixth rule, the critical length becomes

$$l_{\text{crit}} = \frac{t_r}{6 \cdot 6.5\,\text{ps/mm}} \approx 15\,\text{mm}\ (t_r = 0.6\,\text{ns})\ \ldots\ 64\,\text{mm}\ (t_r = 2.5\,\text{ns}),$$

  so the 112 mm lines in star topology are transmission lines.

- **Worst case.** The driver model without one-shot has the fastest edge; the real TXB0108 returns to its 4 kΩ output after the one-shot, which damps the ringing.

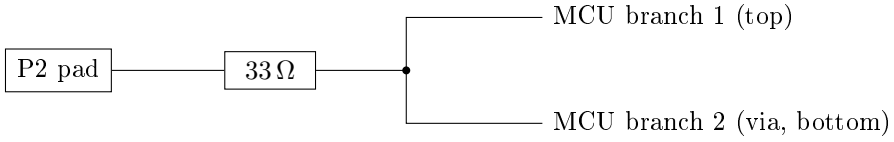
- **Not modelled:** the track of the Jetson carrier between the TXB0108 and the header, the header itself and any losses of the dielectric (the simulation is lossless). All of them reduce the ringing. The **40-way ribbon cable** between the Jetson and the board is not modelled either and can change the result substantially: SCK must therefore be measured on the prototype with the final cable (Section 5).

- **IBIS accuracy:** TI states that IBIS cannot describe the dynamic impedance of the TXB0108; the results are an order of magnitude, not a measurement.

- **SPI0** was not simulated (its pins on P2 are still Passive). Its topology is the same star at P2, so the conclusions are applied to SPI0 by similarity.

3.8 Design Decision

One 33 Ω **series resistor** has been added on each SCK line (SPI0_SCK at P2 pin 23, SPI1_SCK at P2 pin 13), on the top layer, within 3–5 mm of the pad of P2 and **before** the branch point:



- a source termination works only close to the driver; P2 is the closest point available on the board, and it is where Altium placed the resistor in the sweep;
- on the layout the branch point is a via next to the resistor: the top branch leaves the resistor pad, the bottom branch leaves the via (a via connects both layers at the same point);
- 33 Ω is the same part as R4: no new BOM line and no extra JLCPCB setup fee (Basic part);
- $33\,\Omega \times 40\,\text{pF} \approx 1.3\,\text{ns}$: negligible at 1 MHz to 4 MHz;
- damping the reflection that returns to the TXB0108 also protects its one-shot from being re-triggered, a known issue of this translator on long lines;
- MOSI (data, sampled after the ringing) and NSS (stays below V_{IL} after the falling edge, to be confirmed) do not need a resistor; one per MOSI line can be added if space allows.

4 Changes Made to the Project for the Simulations

Where	Change
Schematic, P2	electrical type of pins 13, 16, 18, 22, 37 changed from Passive to I/O (local change of the placed part; an *Update From Library* restores Passive)
Schematic, U1 and P2	SI model parameters (IBIS models per pin) written with Update Models in Schematic
PCB rules	Supply Nets (6 rules), Signal Top Value 3.3 V, Signal Base Value rule deleted; 5V_SPI temporarily set to 3.3 V and restored to 5 V
Schematic, U1 and P2 (SI models)	Technology LVC instead of HC
Schematic and PCB	two 33 Ω series resistors on SPI0_SCK and SPI1_SCK (design change resulting from the simulation)
Documentation	CSV exports of the waveforms in `Documentation\SI_Data`, plotting script `plot_si.py`
Layer Stack Manager	impedance profiles S50 and D90 (USB 90R)
Project	new Draftsman document `BYTE_SPIne.PCBDwf`; folder `SI_Models` with the IBIS files

Table 15: Changes. Apart from the two resistors, none of them affects copper, DRC or BOM; the SI rules are not checked by the DRC unless enabled.

5 Open Points

1. Repeat SCK with the realistic driver model `TXB0108_IOB_OS_33_tp_out` on P2-13.
2. Reflection analysis of NSS3/NSS4, in particular the falling edge (slave selection).
3. Crosstalk SCK → NSS: select both nets, right-click on SCK, Set Aggressor, Crosstalk Waveforms; compare with the estimate of 10 mV to 15 mV of the main documentation [1] (Section 9.5).
4. Repeat the SCK analysis of SPI0 on the layout with its resistor (pins of P2 set to I/O as for SPI1).
5. Bench test on the prototype: SCK at both MCUs with an oscilloscope (probe with short ground spring), with the final ribbon cable, and the SPI test of 10 min listed in the main documentation [1] (Section 11).

6 Quick Reproduction Checklist

1. Stack-up as Figure 1, impedance profile D90 (Section 2).
2. IBIS files in `SI_Models`: `stm32f0_448_lqfp48.ibs`, `txb0108.ibs`.
3. P2 pins 13, 16, 18, 22, 37 electrical type I/O.
4. Tools » Signal Integrity » Model Assignments: U1 and P2 as in the tables above, Update Models in Schematic.
5. Technology LVC on U1 and P2. Rules: Supply Nets, Signal Top Value 3.3 V, no Signal Base Value rule.
6. Reanalyze Design; check Top Value ≈3.3 V on the SPI1 nets.
7. Net to the right list, directions as Table 11, Reflection Waveforms; Serial Res (10–56 Ω, 10 steps) for the sweep.
8. Export: click in a plot, File » Export » Plot, CSV; figures with `python plot_si.py SI_Data SI_Figures`.

References

[1] J. Luca, *Architecture Design and Sizing of the BYTE SPIne SPI-to-CAN Bridge*, BARK Project, October 2026, `Documentation/BYTE_SPIne_Documentation.pdf`.

[2] STMicroelectronics, *STM32F0 IBIS models*, file `stm32f0_448_lqfp48.ibs`, rev. 1.0, product page (*CAD Resources*): https://www.st.com/en/microcontrollers-microprocessors/stm32f072c8.html.

[3] Texas Instruments, *TXB0108 IBIS model*, file `txb0108.ibs`, rev. 1.7, product page (*Design & development*): https://www.ti.com/product/TXB0108.

[4] Texas Instruments, *TXB0108 8-Bit Bidirectional Voltage-Level Translator with Auto-Direction Sensing*, datasheet, https://www.ti.com/product/TXB0108.

[5] IPC, *IPC-2141A Design Guide for High-Speed Controlled Impedance Circuit Boards*; paid standard, https://www.ipc.org.

[6] Altium, *Signal Integrity Analysis (by Altium)*, https://www.altium.com/documentation/altium-designer/analyzing-pcb/si-analysis/si-analyzer-altium.

[7] Altium, *Defining the Layer Stack*, https://www.altium.com/documentation/altium-designer/pcb/defining-layer-stack.

[8] Altium, *Draftsman Transmission Line Table*, https://altium.com/documentation/altium-nexus/draftsman-transmission-line-table.

[9] JLCPCB, *Impedance calculator*, https://jlcpcb.com/capabilities/impedance-calculator.